

SVR R2_Score values for 50 startup dataset:

#	C	gamma	#	Linear R2_Score	#	Poly_R2_score	#	RBF_R2_score	#	Sigmoid_R2score
10	scale			-0.0396449		-0.0536672		-0.056807		-0.0547195
10	auto			-0.0396449		-0.053667		-0.056807		-0.054719
100	scale			0.1064681		-0.0198021		-0.050726		-0.030453
100	auto			0.1064681		-0.0198021		-0.050726		-0.0304535
500	scale			0.5928977		0.114684		-0.024323		0.07057214
500	auto			0.5928977		0.1146848		-0.0243233		0.070572
1000	scale			0.7802839		0.2661637		0.006768		0.185068
1000	auto			0.780283		0.266163		0.006768		0.18506
2000	scale			0.87677		0.481002		0.067515		0.39706
2000	auto			0.87677		0.481002		0.06751		0.39706
3000	scale			0.89567		0.637006		0.123227		0.59136
3000	auto			0.895674		0.637006		0.123227		0.591363
4000	scale			0.89723		0.732637		0.17238		0.628237
4000	auto			0.89723		0.73263		0.172383		0.628237
5000	scale			0.900376		0.79365		0.21242		0.730656
5000	auto			0.900376		0.793655		0.212428		0.730656
6000	scale			0.918187		0.822622		0.24523		0.797246

The SVR has the R2_score value= 0.918187

Decision Tree Regressor:

Criterion	Splitter	max_features	r2_score
squared_error	best	None	0.91431
squared_error	random	None	0.73062
squared_error	best	sqrt	0.75753
squared_error	best	log2	0.31687
squared_error	random	sqrt	0.58516
squared_error	random	log2	-0.08104
friedman_mse	best	None	0.92008
friedman_mse	random	None	0.91731
friedman_mse	best	sqrt	0.61627
friedman_mse	best	log2	0.68112
friedman_mse	random	sqrt	0.52389
friedman_mse	random	log2	0.22149
absolute_error	best	None	0.94855
absolute_error	random	None	0.79868
absolute_error	best	sqrt	0.59529
absolute_error	best	log2	0.840729
absolute_error	random	sqrt	-0.17268
absolute_error	random	log2	0.16819
poisson	best	None	0.93268
poisson	random	None	0.938906
poisson	best	sqrt	0.43485
poisson	best	log2	0.64464
poisson	random	sqrt	0.52958
poisson	random	log2	0.835202

The Decision Tree Regressor has the R2_score value=> 0.938906

By comparing the both , The Decision tree model performed well.